Haotong Hu

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EDUCATION

- UCLA | Bachelor of Computer Science | *2025 - 2029* (*Expected*)

TECHNICAL PROJECTS

AI Model Training and RL | Lemur Lab at UCLA | *Mar 2026 – Sep 2026*

- Responsible for Real-time Sign Language Translation project, conducted **training** of Uni-Sign and mT5-base models on **GPU** (@ NVIDIA Titan RTX & DGX Spark/GB10 with CUDA 12.2).
- Validate Reinforcement Learning **GRPO Reward Model** based on Semantic Weight of words and Source-Target Alignment Strength for Credit Assignment
- **Fine-tuned** model with How2Sign dataset, completed 10 comparative experiments to analyze cross-modal alignment, information missing and etc, achieved peak BLEU-4 score of 14.43.
- Validated different **DPO (Direct Preference Optimization)** training methods on impact of incorporating context into different training methods, achieved peak BLEU-4 score of 23.05.
- Developed infra tools for idle GPU detection, auto-launch, early stopping, and fault recovery to support multi-day training; solved video loading anomalies and distributed training deadlocks.

Agent Platform Dev / Intern | **Shenzhen Agilean Software Co., Ltd.** | *July 2026 – Oct 2026*

- Designed and implemented a Jenkins Inbound Agent like architecture in agile R&D AI Agent platform, upgraded from local config to flexible scheduling mode, established end-to-end pipeline for multi-worker registration, environment provisioning, session scheduling, etc.
- Built Plan-to-Repo dev framework to seamlessly integrate AI capabilities into existing workflows, enabling continuous updates on progress, decision-making, and result validation.
- Successfully verified software projects such as Frappe/Lending management system, delivering high-quality solutions to premium enterprise customers including Fullgoal Fund and CGN.

RESEARCH & PUBLICATIONS

- **Publication:** *A Self-Reflective RAG to Eliminate Hallucination in LLM* (Published in CSIT 2024).
- **Patent: Lead Inventor**, Self-RAG Algorithm Patent (#63/663,192).

PROFESSIONAL EXPERIENCE & LEADERSHIP

- **ACM AI Committee (UCLA) | Operation Director** | *Mar 2026 – Now*
- Organized flagship events, including co-hosted ACM AI x Google Gemini workshop, "AI in Practice" joint event with Product Space, full-day AI hackathon with 70+ attendees, etc.

TECHNICAL SKILLS

- **AI/Machine Learning:** AI Model Training, Reinforcement Learning, NLP, RAG, CNN/RNN.
- **Languages & Framework:** PyTorch, DeepSpeed, Python, TypeScript, C++, Node.JS, React
- **Development Tools:** Deep user of Claude Code, Codex, VS Code, Git, and etc.